



PROOF OF CONCEPT · PREPARED FOR ATHA GROUP

CAPEX & WBS Control Hub

Budget control from CAPEX approval to asset capitalisation

Commitment control

Open POs count against budget, not just bills

Multi-level WBS

Project, sub-project and work-package rollup

Zoho ERP integration

Grounded in Zoho's own API specification

01 The control gap today
Why budget monitored against bills alone is monitored too late

02 The control principle
One formula that governs every screen in the application

03 The working application
Eleven screens, walked through with live screens

04 Preventing double counting
How a purchase order can never be counted twice

05 Zoho ERP integration
What is possible, and what the API genuinely cannot do

06 What this proves, and what comes next
Scope of the proof of concept and the road to production

A budget checked only against vendor bills is checked too late.

By the time a vendor bill arrives, the money is already spent. The purchase order committed it weeks earlier. Any control that looks only at actual cost is reporting history, not preventing overrun.

Without commitment control

- Budget looks available because bills have not arrived
- A second purchase order is approved against money already promised
- Overrun is discovered at invoice, when it can no longer be prevented
- CWIP position is understated until month-end
- Overruns are settled in conversation, leaving no record of who approved what, or why

With the Control Hub

- Approved purchase orders consume budget the moment they are raised
- Availability is tested before approval, at the work-package level
- Over-budget requests route to exception approval, with a recorded reason
- Received-but-unbilled value is visible as exposure, not hidden
- Every override is captured with its reason, approver and timestamp

The difference is timing, and timing is the whole control.

One approach reports an overrun after it has happened. The other prevents it at the point of approval.

THE CONTROL PRINCIPLE

$$\text{Available Budget} = \text{Current Approved Budget} - \text{Exposure}$$

$$\text{Exposure} = \text{Open PO Commitment} + \text{Actual CWIP}$$

Taken from Atha Group's own requirement document. Every figure in that document was re-verified line by line.

What follows from it

The original budget is never overwritten

Supplements, returns and transfers are recorded as separate versioned lines, each carrying requestor, approver, date, reason and approval reference. The figure first approved stays visible for ever.

Budget is controlled where it is approved

Spending posts on a work package while budget sits above it. The check resolves to the nearest element carrying budget and evaluates its whole subtree, so a parent cannot look untouched while its children spend.

Thresholds are explicit

Warning at 80 percent. Exception approval at 90 percent. A hard stop once exposure exceeds the approved budget, with the shortfall stated in rupees.

WORKED EXAMPLE, FROM ATHA GROUP'S OWN REQUIREMENT DOCUMENT

$$\text{₹80,00,000 budget} - \text{₹35,00,000 actual CWIP} - \text{₹30,00,000 open commitment} = \text{₹15,00,000 available}$$

A proposed purchase order of ₹20,00,000 exceeds that by ₹5,00,000, so it routes to exception approval or budget revision rather than being treated as a within-budget order.

Every project's true position, on one screen

Management sees what has been committed, not only what has been spent.

CAPEX & WBS Control Hub

All entities

All plants

Compact

Approvals 4

Alerts 4

NR

WORK

Executive Dashboard

My Approvals 4

Alerts & Exceptions 4

PROJECT CONTROL

CAPEX Projects

WBS Explorer

Budget Planning Grid

Budget Availability Check

Budget Revisions

PROCUREMENT & ACTUALS

Purchase Requests

Commitments (PO)

GRN & Receipts

Vendor Bills & CWIP

Commitment Reconciliation

CLOSURE

Capitalisation

INTEGRATION

Zoho ERP Connector

API Inventory

GOVERNANCE

Audit Trail

Home > Executive Dashboard

Executive CAPEX Dashboard

Portfolio budget

₹2.90 Cr

Exposure

₹1.80 Cr

Available

₹1.09 Cr

Utilisation

62.2%

CURRENT APPROVED BUDGET

₹2.90 Cr

3 active projects

OPEN PO COMMITMENT

₹1.01 Cr

unbilled balance of approved POs

ACTUAL CWIP

₹79.8L

posted vendor bills

AVAILABLE BUDGET

₹1.09 Cr

62.2% utilised

RECEIVED, NOT BILLED

₹5.0L

exposure awaiting vendor bill

Budget vs Commitment vs Actual vs Available

click a row to open the WBS hierarchy

CAPEX Code	Project	Plant	Status	Current Budget	Commitment	Actual CWIP	Available	Utilisation	Exposure
CAPEX-2026-001	New Production Line	Angul Integrated Plant	Released	₹1,60,00,000	₹55,50,000	₹52,00,000	₹52,50,000	67.2%	
CAPEX-2026-002	Warehouse Expansion	Angul Integrated Plant	Released	₹80,00,000	₹25,20,000	₹16,80,000	₹38,00,000	52.5%	
CAPEX-2026-003	Solar Power Project	Dhenkanal Unit II	Released	₹50,00,000	₹20,00,000	₹11,00,000	₹19,00,000	62.0%	
Portfolio total				₹2,90,00,000	₹1,00,70,000	₹79,80,000	₹1,09,50,000	62.2%	

Actual CWIP

Open commitment

Available

Exceptions requiring attention

PR-2026-0010

Available budget insufficient. Routed for exception approval per PDF section 7.MSG-BUD-001

PR-2026-0021

The proposed value of ₹25,00,000 exceeds the available budget of ₹19,10,000 for CAPEX-2026-001.03.01 (budget is held at CAPEX-2026-001.03) (Plant & Machinery) by ₹5,90,000. Submit a budget revision or request exception approval.

Bill exceeds purchase order onPO-2026-0008(CAPEX-2026-001.05): billed ₹7,60,000 against an ordered value of ₹7,00,000.MSG-PRC-004

CAPEX-2026-001.03.02

has ₹5,00,000 received but not yet billed on PO-2026-0005. This is included in exposure and converts to actual CWIP when the vendor bills.MSG-PRC-005

Budget revisionREV-003(SUPPLEMENT) is awaiting approval — Additional instrumentation scope identified during commissioning readiness review.

- Budget, commitment, actual and available for the whole portfolio
- Utilisation banded green, amber and red against the approved budget
- Exceptions surfaced immediately, not buried in a report
- Click any project to open its work breakdown structure

Multi-level work breakdown, with figures that roll up

The client document has no WBS layer at all. This is the structure that makes project control possible.

CAPEX & WBS Control Hub | All entities | All plants | Compact | Approvals 0 | Alerts 0 | NR

WORK

- Executive Dashboard
- My Approvals
- Alerts & Exceptions

PROJECT CONTROL

- CAPEX Projects
- WBS Explorer**
- Budget Planning Grid
- Budget Availability Check
- Budget Revisions

PROCUREMENT & ACTUALS

- Purchase Requests
- Commitments (PO)
- GRN & Receipts
- Vendor Bills & CWIP
- Commitment Reconciliation

CLOSURE

- Capitalisation

INTEGRATION

- Zoho ERP Connector
- API Inventory

GOVERNANCE

- Audit Trail

Home > CAPEX Projects > CAPEX-2026-001

WBS Hierarchy — New Production Line

CAPEX code: CAPEX-2026-001 | Plant: Angul Integrated Plant | Status: ● Released | Current budget: ₹1,60,00,000 | Exposure: ₹1,07,50,000 | Available: ₹52,50,000 | Utilisation: 67.2%

Project: CAPEX-2026-001 — New Production Line

Parent rows show the rolled-up total of the element and every descendant.

WBS Element	Description	Budget Head	Status	Progress	Budget	Commitment	Actual CWIP	Recd-Unbilled	Available	Util.	Exposure	Pr
CAPEX-2026-001.01	Land and Site Development	Land & Site Development	● Technically Completed	100%	no budget	₹0	₹0	₹0	—	—		●
▼ CAPEX-2026-001.02	Civil and Structural Work	Civil	● Released	62%	₹25,00,000	₹3,00,000	₹11,00,000	₹0	₹11,00,000	56.0%		●
▼ CAPEX-2026-001.02.01	Foundation and Piling	Civil	● Released	90%	at CAPEX-2026-001.02	₹3,00,000	₹11,00,000	₹0	—	—		●
CAPEX-2026-001.02.01.01	Site Piling Works	Civil	● Released	100%	at CAPEX-2026-001.02	₹0	₹8,00,000	₹0	—	—		●
CAPEX-2026-001.02.01.02	Foundation Concreting	Civil	● Released	80%	at CAPEX-2026-001.02	₹3,00,000	₹3,00,000	₹0	—	—		●
CAPEX-2026-001.02.02	Structural Steel Erection	Civil	● Released	45%	at CAPEX-2026-001.02	₹0	₹0	₹0	—	—		●
▼ CAPEX-2026-001.03	Plant and Machinery	Plant & Machinery	● Released	48%	₹90,00,000	₹46,50,000	₹24,40,000	₹5,00,000	₹19,10,000	78.8%		●
CAPEX-2026-001.03.01	Rolling Mill Equipment	Plant & Machinery	● Released	55%	at CAPEX-2026-001.03	₹39,00,000	₹17,40,000	₹0	—	—		●
CAPEX-2026-001.03.02	Material Handling System	Plant & Machinery	● Released	40%	at CAPEX-2026-001.03	₹7,50,000	₹7,00,000	₹5,00,000	—	—		●
CAPEX-2026-001.04	Electrical	Electrical	● Released	35%	₹23,00,000	₹3,00,000	₹6,00,000	₹0	₹14,00,000	39.1%		●
CAPEX-2026-001.05	Installation	Installation	● Released	20%	₹12,00,000	₹0	₹7,60,000	₹0	₹4,40,000	63.3%		●
CAPEX-2026-001.06	Consultancy and Project Management	Consultancy & Other	● Released	55%	₹10,00,000	₹3,00,000	₹3,00,000	₹0	₹4,00,000	60.0%		●
CAPEX-2026-001.07	Instrumentation and Automation	Instrumentation & Automation	● Released	10%	no budget	₹0	₹0	₹0	—	—		●
CAPEX-2026-001.08	Pre-operative Expenses	Consultancy & Other	● Released	30%	no budget	₹0	₹0	₹0	—	—		●
CAPEX-2026-001.09	Effluent Treatment - Phase 2 (abandoned)	Utilities	● Closed	15%	no budget	₹0	₹0	₹0	—	—		●
Project total					₹1,60,00,000	₹55,50,000	₹52,00,000	₹5,00,000	₹52,50,000	67.2%		

● Three levels: project, sub-project and work package

● Parents show the total of themselves and every element beneath

● Budget, commitment, actual, received-not-billed and available per element

● Elements that cannot be procured against are marked clearly

Original budget and revisions, kept visibly separate

Auditors can always see what was first approved, and every change made since.

☰

CAPEX & WBS Control Hub

All entities

All plants

☰ Compact

✓ Approvals 0

⚠ Alerts 0

NR

WORK

☑ Executive Dashboard

✓ My Approvals

⚠ Alerts & Exceptions

PROJECT CONTROL

☰ CAPEX Projects

🔗 WBS Explorer

■ Budget Planning Grid

⦿ Budget Availability Check

🔄 Budget Revisions

PROCUREMENT & ACTUALS

🔍 Purchase Requests

📄 Commitments (PO)

📉 GRN & Receipts

₹ Vendor Bills & CWIP

⚖ Commitment Reconciliation

CLOSURE

★ Capitalisation

INTEGRATION

🔗 Zoho ERP Connector

Home > Budgets

Budget Planning Grid

Original
₹1,50,00,000

Revisions
₹10,00,000

Current approved
₹1,60,00,000

Available
₹52,50,000

Project

CAPEX-2026-001 — New Production Line

Budget by WBS element and budget head

WBS	Description	Budget Head	Original	Revisions	Current Approved	PO Commitment	Actual CWIP	Available	Exposure %
CAPEX-2026-001.02	Civil and Structural Work	Civil	₹25,00,000	₹0	₹25,00,000	₹0	₹0	₹25,00,000	0.0%
CAPEX-2026-001.02.01.01	Site Piling Works	Civil	₹0	₹0	₹0	₹0	₹8,00,000	(₹8,00,000)	0.0%
CAPEX-2026-001.02.01.02	Foundation Concreting	Civil	₹0	₹0	₹0	₹3,00,000	₹3,00,000	(₹6,00,000)	0.0%
CAPEX-2026-001.03	Plant and Machinery	Plant & Machinery	₹80,00,000	₹10,00,000	₹90,00,000	₹0	₹0	₹90,00,000	0.0%
CAPEX-2026-001.03.01	Rolling Mill Equipment	Plant & Machinery	₹0	₹0	₹0	₹39,00,000	₹17,40,000	(₹56,40,000)	0.0%
CAPEX-2026-001.03.02	Material Handling System	Plant & Machinery	₹0	₹0	₹0	₹7,50,000	₹7,00,000	(₹14,50,000)	0.0%
CAPEX-2026-001.04	Electrical	Electrical	₹20,00,000	₹3,00,000	₹23,00,000	₹3,00,000	₹6,00,000	₹14,00,000	39.1%
CAPEX-2026-001.05	Installation	Installation	₹15,00,000	(₹3,00,000)	₹12,00,000	₹0	₹7,60,000	₹4,40,000	63.3%
CAPEX-2026-001.06	Consultancy and Project Management	Consultancy & Other	₹10,00,000	₹0	₹10,00,000	₹3,00,000	₹3,00,000	₹4,00,000	60.0%
Total			₹1,50,00,000	₹10,00,000	₹1,60,00,000	₹55,50,000	₹52,00,000	₹52,50,000	67.2%

• The original approved budget is immutable. Supplements, returns and transfers are recorded as separate versioned lines, each carrying requestor, approver, date, reason and approval reference.

- Original, revisions and current approved budget shown side by side
- Commitment, actual and available against each budget head
- Exposure percentage per element
- The original figure is never altered by a revision

The control, exercised before money is committed

This is the moment of control. The message states the shortfall in rupees and offers the route forward.

CAPEX & WBS Control Hub

All entitiesAll plants

CompactApprovals 0Alerts 0NR

WORK

Executive DashboardMy ApprovalsAlerts & Exceptions

PROJECT CONTROL

CAPEX ProjectsWBS ExplorerBudget Planning Grid

Budget Availability Check

Budget Revisions

PROCUREMENT & ACTUALS

Purchase RequestsCommitments (PO)GRN & ReceiptsVendor Bills & CWIPCommitment Reconciliation

CLOSURE

Capitalisation

INTEGRATION

Zoho ERP Connector

Home > Budgets > Availability Check

Budget Availability Check

Simulate a proposed commitment

Project

CAPEX-2026-001 — New Production Line

WBS element

CAPEX-2026-001.03 — Plant and Machinery

Proposed value (₹)

2000000

Run check

✖ The proposed value of ₹20,00,000 exceeds the available budget of ₹19,10,000 for CAPEX-2026-001.03 (Plant & Machinery) by ₹90,000. Submit a budget revision or request exception approval.

MSG-BUD-001

Submit budget revision

Request exception approval

Position at CAPEX-2026-001.03 (rolled up)

Current approved budget	₹90,00,000
Open PO commitment	₹46,50,000
Actual CWIP	₹24,40,000
Received not billed	₹5,00,000
Exposure	₹70,90,000
Available before	₹19,10,000
Proposed	₹20,00,000
Available after	(₹90,000)

How the check works

Exposure

Open PO Commitment + Actual CWIP

Available Budget

Current Approved Budget – Exposure

Utilisation

Exposure ÷ Current Approved Budget

Budget held on a WBS element covers that element and every descendant, so the check evaluates the rolled-up position. Checking only the element's own line would let a parent's budget look untouched while its children consumed it.

Thresholds: warning at 80%, exception approval at 90%, hard stop when exposure exceeds the approved budget.

- Simulate any proposed commitment before raising it

● Full position shown: budget, commitment, actual, exposure, available

● Message names the shortfall and the next action

● Deep-linkable, so a specific check can be shared for approval

Every request budget-checked at submission

Overrun becomes a governed decision with a named approver, instead of an accident.

CAPEX & WBS Control Hub

All entities

All plants

Compact

Approvals 0

Alerts 0

NR

WORK

Executive Dashboard

My Approvals

Alerts & Exceptions

PROJECT CONTROL

CAPEX Projects

WBS Explorer

Budget Planning Grid

Budget Availability Check

Budget Revisions

PROCUREMENT & ACTUALS

Purchase Requests

Commitments (PO)

GRN & Receipts

Vendor Bills & CWIP

Commitment Reconciliation

CLOSURE

Capitalisation

INTEGRATION

Zoho ERP Connector

API Inventory

GOVERNANCE

Audit Trail

Home

Purchase Requests

Purchase Requests

Create purchase request

Purchase request control view

PR Number	Project	WBS	Budget Head	Description	Value	Budget Check	Status	Approver
PR-2026-0001	CAPEX-2026-001	CAPEX-2026-001.02.01.01	Civil	Piling contractor - 120 piles	₹8,00,000	Within budget	Approved	U-PLH
PR-2026-0002	CAPEX-2026-001	CAPEX-2026-001.02.01.02	Civil	RCC foundation works	₹6,00,000	Within budget	Approved	U-PLH
PR-2026-0003	CAPEX-2026-001	CAPEX-2026-001.03.01	Plant & Machinery	Rolling mill main drive	₹30,00,000	Within budget	Approved	U-CFO
PR-2026-0004	CAPEX-2026-001	CAPEX-2026-001.03.01	Plant & Machinery	Mill stands and guides	₹22,00,000	Within budget	Approved	U-CFO
PR-2026-0005	CAPEX-2026-001	CAPEX-2026-001.03.02	Plant & Machinery	Overhead crane 20T	₹12,00,000	Within budget	Approved	U-CFO
PR-2026-0006	CAPEX-2026-001	CAPEX-2026-001.04	Electrical	HT panels and switchgear	₹9,00,000	Within budget	Approved	U-PLH
PR-2026-0007	CAPEX-2026-001	CAPEX-2026-001.04	Electrical	Cabling and trays	₹5,00,000	Within budget	Approved	U-PLH
PR-2026-0008	CAPEX-2026-001	CAPEX-2026-001.05	Installation	Mechanical erection contractor	₹7,00,000	Within budget	Approved	U-PLH
PR-2026-0009	CAPEX-2026-001	CAPEX-2026-001.06	Consultancy & Other	Detailed engineering consultancy	₹6,00,000	Within budget	Approved	U-FIN
PR-2026-0010	CAPEX-2026-001	CAPEX-2026-001.03.02	Plant & Machinery	Conveyor system - additional scope	₹20,00,000	Exceeds budget	Exception Pending	Approve
PR-2026-0011	CAPEX-2026-002	CAPEX-2026-002.01	Civil	Warehouse shed civil	₹28,00,000	Within budget	Approved	U-PLH
PR-2026-0012	CAPEX-2026-002	CAPEX-2026-002.02	Plant & Machinery	Pallet racking system	₹14,00,000	Within budget	Approved	U-PLH
PR-2026-0013	CAPEX-2026-003	CAPEX-2026-003.01	Plant & Machinery	Solar modules 2 MW	₹22,00,000	Within budget	Approved	U-CFO
PR-2026-0014	CAPEX-2026-003	CAPEX-2026-003.02	Electrical	Inverters and transformers	₹9,00,000	Within budget	Approved	U-PLH
PR-2026-0015	CAPEX-2026-001	CAPEX-2026-001.02.02	Civil	Structural steel supply	₹9,00,000	Within budget	Submitted	Approve
PR-2026-0016	CAPEX-2026-001	CAPEX-2026-001.07	Instrumentation & Automation	PLC and SCADA package	₹4,00,000	Within budget	Submitted	Approve
PR-2026-0017	CAPEX-2026-001	CAPEX-2026-001.06	Consultancy & Other	Third-party inspection	₹1,50,000	not run	Draft	
PR-2026-0018	CAPEX-2026-002	CAPEX-2026-002.01	Civil	Site drainage	₹2,00,000	not run	Draft	
PR-2026-0019	CAPEX-2026-003	CAPEX-2026-003.01	Plant & Machinery	Module mounting structures	₹5,00,000	Within budget	Submitted	Approve
PR-2026-0020	CAPEX-2026-001	CAPEX-2026-001.09	Utilities	ETP Phase 2 civil (project abandoned)	₹3,00,000	not run	Rejected	U-PLH
PR-2026-0021	CAPEX-2026-001	CAPEX-2026-001.03.01	Plant & Machinery	Extra mill scope	₹25,00,000	Exceeds budget	Exception Pending	Approve

● Availability check runs automatically when a request is submitted

● Within-budget requests follow the normal approval route

● Over-budget requests route to exception approval, never silently blocked

● An exception cannot be approved without a recorded reason

Purchase orders as committed expenditure, line by line

Commitment is tracked at line level, which is what makes work-package control possible.

WORK

EXECUTIVE DASHBOARD

MY APPROVALS

ALERTS & EXCEPTIONS

PROJECT CONTROL

CAPEX PROJECTS

WBS EXPLORER

BUDGET PLANNING GRID

BUDGET AVAILABILITY CHECK

BUDGET REVISIONS

PROCUREMENT & ACTUALS

PURCHASE REQUESTS

COMMITMENTS (PO)

GRN & RECEIPTS

VENDOR BILLS & CWIP

COMMITMENT RECONCILIATION

CLOSURE

CAPITALISATION

INTEGRATION

ZOHO ERP CONNECTOR

API INVENTORY

GOVERNANCE

AUDIT TRAIL

CAPEX & WBS CONTROL HUB

ALL ENTITIES

ALL PLANTS

COMPACT

APPROVALS

ALERTS

NR

Home > Commitments

Commitments — Purchase Orders

Purchase order commitment view

PO Number	Vendor	Project	Status	Curr.	Ordered	Billed	Open Commitment	Amend	Actions
PO-2026-0001	Odisha Piling Works Pvt Ltd	CAPEX-2026-001	Fully Actualised	INR	₹8,00,000	₹8,00,000	₹0	—	Amend Cancel Close
	Line 1 · CAPEX-2026-001.02.01.01 · Piling - 120 nos				₹8,00,000	₹8,00,000	₹0	Fully billed	
PO-2026-0002	Konark Constructions	CAPEX-2026-001	Partially Actualised	INR	₹6,00,000	₹3,00,000	₹3,00,000	—	Amend Cancel Close
	Line 2 · CAPEX-2026-001.02.01.02 · RCC foundation				₹6,00,000	₹3,00,000	₹3,00,000	Partially billed	
PO-2026-0003	Bharat Heavy Engineering Co	CAPEX-2026-001	Partially Actualised	INR	₹34,40,000	₹17,40,000	₹17,00,000	1	Amend Cancel Close
	Line 3 · CAPEX-2026-001.03.01 · Rolling mill main drive				₹34,40,000	₹17,40,000	₹17,00,000	Partially billed	
PO-2026-0004	Bharat Heavy Engineering Co	CAPEX-2026-001	Fully Committed	INR	₹22,00,000	₹0	₹22,00,000	—	Amend Cancel Close
	Line 4 · CAPEX-2026-001.03.01 · Mill stands and guides				₹22,00,000	₹0	₹22,00,000	PO approved, not billed	
PO-2026-0005	Eastern Cranes Ltd	CAPEX-2026-001	Partially Actualised	INR	₹12,50,000	₹7,00,000	₹5,50,000	—	Amend Cancel Close
	Line 5 · CAPEX-2026-001.03.02 · Overhead crane 20T				₹12,50,000	₹7,00,000	₹5,50,000	Partially billed	
PO-2026-0006	Utkal Electricals	CAPEX-2026-001	Partially Actualised	INR	₹9,00,000	₹6,00,000	₹3,00,000	—	Amend Cancel Close
	Line 6 · CAPEX-2026-001.04 · HT panel				₹6,00,000	₹6,00,000	₹0	Fully billed	
	Line 7 · CAPEX-2026-001.04 · LT switchgear				₹3,00,000	₹0	₹3,00,000	PO approved, not billed	
PO-2026-0007	Utkal Electricals	CAPEX-2026-001	Cancelled	INR	₹5,00,000	₹0	₹0	—	released
	Line 8 · CAPEX-2026-001.04 · Cabling and trays				₹5,00,000	₹0	₹0	PO cancelled before billing	
PO-2026-0008	Sriram Erectors	CAPEX-2026-001	Fully Committed	INR	₹7,00,000	₹7,60,000	₹0	—	Amend Cancel Close
	Line 9 · CAPEX-2026-001.05 · Mechanical erection				₹7,00,000	₹7,60,000	₹0	Bill exceeds PO	
PO-2026-0009	Delta Engineering Consultants	CAPEX-2026-001	Partially Actualised	INR	₹6,00,000	₹3,00,000	₹3,00,000	—	Amend Cancel Close
	Line 10 · CAPEX-2026-001.06 · Detailed engineering				₹6,00,000	₹3,00,000	₹3,00,000	Partially billed	
PO-2026-0010	Konark Constructions	CAPEX-2026-002	Partially Actualised	INR	₹28,00,000	₹16,80,000	₹11,20,000	—	Amend Cancel Close
	Line 11 · CAPEX-2026-002.01 · Warehouse shed civil				₹28,00,000	₹16,80,000	₹11,20,000	Partially billed	
PO-2026-0011	Storefit Systems	CAPEX-2026-002	Fully Committed	INR	₹14,00,000	₹0	₹14,00,000	—	Amend Cancel Close
	Line 12 · CAPEX-2026-002.02 · Pallet racking				₹14,00,000	₹0	₹14,00,000	PO approved, not billed	
PO-2026-0012	SunPeak Energy GmbH	CAPEX-2026-003	Partially Actualised	EUR @92.5	₹22,00,000	₹11,00,000	₹11,00,000	—	Amend Cancel Close
	Line 13 · CAPEX-2026-003.01 · Solar modules 2 MW				₹22,00,000	₹11,00,000	₹11,00,000	Partially billed	
PO-2026-0013	Utkal Electricals	CAPEX-2026-003	Fully Committed	INR	₹9,00,000	₹0	₹9,00,000	—	Amend Cancel Close
	Line 14 · CAPEX-2026-003.02 · Inverters and transformers				₹9,00,000	₹0	₹9,00,000	PO approved, not billed	
PO-2026-0014	Odisha Piling Works Pvt Ltd	CAPEX-2026-001	Closed	INR	₹2,00,000	₹0	₹0	—	released
	Line 15 · CAPEX-2026-001.02.01.01 · Additional piling - provisional				₹2,00,000	₹0	₹0	PO closed - residual released	
PO-2026-0015	Eastern Cranes Ltd	CAPEX-2026-001	Fully Committed	INR	₹2,00,000	₹0	₹2,00,000	—	Amend Cancel Close
	Line 16 · CAPEX-2026-001.03.02 · Crane rails and stops				₹2,00,000	₹0	₹2,00,000	PO approved, not billed	

- Ordered, billed and open commitment for every order and every line
- Amendment revalidates the incremental value against budget
- Cancellation releases the full commitment; closure releases only the residual
- Foreign-currency orders carry currency and exchange rate

Purchase order to receipt to bill, reconciled per line

For any purchase order, billed plus open commitment always equals the ordered value.

☰

CAPEX & WBS Control Hub

All entities

All plants

Compact

✓ Approvals 0

⚠ Alerts 0

NR

WORK

Executive Dashboard

✓ My Approvals

⚠ Alerts & Exceptions

PROJECT CONTROL

CAPEX Projects

WBS Explorer

Budget Planning Grid

Budget Availability Check

Budget Revisions

PROCUREMENT & ACTUALS

Purchase Requests

Commitments (PO)

GRN & Receipts

Vendor Bills & CWIP

Commitment Reconciliation

CLOSURE

Capitalisation

INTEGRATION

Zoho ERP Connector

API Inventory

GOVERNANCE

Audit Trail

Home > Reconciliation

Commitment-to-Actual Reconciliation

Lines16

Open commitment₹1,00,70,000

Billed₹79,80,000

Received not billed₹5,00,000

Exceptions2

Line-level reconciliation — purchase order → receipt → bill

PO	WBS	Description	Ordered	Received	Billed	Open Commitment	Recd-Unbilled	Exposure	Position
PO-2026-0001	CAPEX-2026-001.02.01.01	Piling - 120 nos	₹8,00,000	₹8,00,000	₹8,00,000	₹0	₹0	₹8,00,000	Fully billed
PO-2026-0002	CAPEX-2026-001.02.01.02	RCC foundation	₹6,00,000	₹3,00,000	₹3,00,000	₹3,00,000	₹0	₹6,00,000	Partially billed
PO-2026-0003	CAPEX-2026-001.03.01	Rolling mill main drive	₹34,40,000	₹17,00,000	₹17,40,000	₹17,00,000	₹0	₹34,40,000	Partially billed
PO-2026-0004	CAPEX-2026-001.03.01	Mill stands and guides	₹22,00,000	₹0	₹0	₹22,00,000	₹0	₹22,00,000	PO approved, not billed
PO-2026-0005	CAPEX-2026-001.03.02	Overhead crane 20T	₹12,50,000	₹12,00,000	₹7,00,000	₹5,50,000	₹5,00,000	₹12,50,000	Partially billed
PO-2026-0006	CAPEX-2026-001.04	HT panel	₹6,00,000	₹6,00,000	₹6,00,000	₹0	₹0	₹6,00,000	Fully billed
PO-2026-0006	CAPEX-2026-001.04	LT switchgear	₹3,00,000	₹0	₹0	₹3,00,000	₹0	₹3,00,000	PO approved, not billed
PO-2026-0007	CAPEX-2026-001.04	Cabling and trays	₹5,00,000	₹0	₹0	₹0	₹0	₹0	PO cancelled before billing
PO-2026-0008	CAPEX-2026-001.05	Mechanical erection	₹7,00,000	₹0	₹7,60,000	₹0	₹0	₹7,60,000	Bill exceeds PO
PO-2026-0009	CAPEX-2026-001.06	Detailed engineering	₹6,00,000	₹3,00,000	₹3,00,000	₹3,00,000	₹0	₹6,00,000	Partially billed
PO-2026-0010	CAPEX-2026-002.01	Warehouse shed civil	₹28,00,000	₹16,80,000	₹16,80,000	₹11,20,000	₹0	₹28,00,000	Partially billed
PO-2026-0011	CAPEX-2026-002.02	Pallet racking	₹14,00,000	₹0	₹0	₹14,00,000	₹0	₹14,00,000	PO approved, not billed
PO-2026-0012	CAPEX-2026-003.01	Solar modules 2 MW	₹22,00,000	₹11,00,000	₹11,00,000	₹11,00,000	₹0	₹22,00,000	Partially billed
PO-2026-0013	CAPEX-2026-003.02	Inverters and transformers	₹9,00,000	₹0	₹0	₹9,00,000	₹0	₹9,00,000	PO approved, not billed
PO-2026-0014	CAPEX-2026-001.02.01.01	Additional piling - provisional	₹2,00,000	₹0	₹0	₹0	₹0	₹0	PO closed - residual released
PO-2026-0015	CAPEX-2026-001.03.02	Crane rails and stops	₹2,00,000	₹0	₹0	₹2,00,000	₹0	₹2,00,000	PO approved, not billed
Total					₹79,80,000	₹1,00,70,000	₹5,00,000		

✓ Anti-double-count proof: for any purchase order, **billed + open commitment = ordered**. Billing moves value from commitment to actual and never adds to both, so total exposure is unchanged by billing progress.

● Every transaction position covered, from not-yet-billed to over-billed

● Received-but-unbilled tracked as its own exposure bucket

● Cancelled and closed orders show the value released

● Exceptions flagged for action, not left to be discovered

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Pg-11

CWIP to fixed asset, with the gate that protects asset value

An asset capitalised while commitments remain open is an asset carried at the wrong value.

☰

CAPEX & WBS Control Hub

All entities

All plants

☰ Compact

✓ Approvals 0

⚠ Alerts 0

NR

WORK

Executive Dashboard

✓ My Approvals

⚠ Alerts & Exceptions

PROJECT CONTROL

CAPEX Projects

WBS Explorer

Budget Planning Grid

Budget Availability Check

Budget Revisions

PROCUREMENT & ACTUALS

Purchase Requests

Home > Capitalisation

Capitalisation Workbench

CAP-2026-0001 — CAPEX-2026-001 New Production Line

Submitted

CWIP balance	₹52,00,000
Open commitment	₹55,50,000
Received not billed	₹5,00,000
Allocated	₹0
Capitalisation date	2026-08-31

! CAPEX-2026-001 has ₹55,50,000 in open commitments and ₹5,00,000 received but unbilled. Resolve or explicitly write off these before capitalising, or the asset value will be understated.MSG-CAP-001

No asset allocations yet.

Allocate to asset

Approve capitalisation

- CWIP balance, open commitment and unbilled value shown before approval

● Allocation to one or many assets, or a recorded write-off

● Approval refused unless allocations equal the CWIP balance exactly

● Project closes to further procurement on capitalisation

Configuration-driven, least-privilege, and honest about limits

Missing scopes are named with their business impact, so the fix is obvious.

CAPEX & WBS Control Hub

All entitiesAll plants

CompactApprovals 0Alerts 0NR

WORK

Executive Dashboard

My Approvals

Alerts & Exceptions

PROJECT CONTROL

CAPEX Projects

WBS Explorer

Budget Planning Grid

Budget Availability Check

Budget Revisions

PROCUREMENT & ACTUALS

Purchase Requests

Commitments (PO)

GRN & Receipts

Vendor Bills & CWIP

Commitment Reconciliation

CLOSURE

Capitalisation

INTEGRATION

Zoho ERP Connector

API Inventory

GOVERNANCE

Audit Trail

Home > Integration > Zoho ERP Connector

Zoho ERP Connector

Authorise

Authorise (withhold 2 scopes)

Refresh token

Run connectivity tests

Connection: Atha Steel - Zoho ERP (Sandbox)

Data centre: IN

OAuth: Connected

Organisation: Atha Steel & Power Ltd

Scopes granted: 44 / 45

Connection profile

Environment: Sandbox

Accounts domain: https://accounts.zoho.in

API domain: https://www.zohoapis.in/erp/v3

Client ID: secretref://kv/zoho/client_id

Client secret: secretref://kv/zoho/client_secret - never displayed after saving

Refresh token: secretref://kv/zoho/refresh_token

Redirect URI: http://localhost:8080/api/zoho/callback

Token expiry: 2026-08-06T05:30:40

Data centres

Region	Accounts domain	API domain	Evidence source
IN	https://accounts.zoho.in	https://www.zohoapis.in	✓ Zoho ERP documentation
US	https://accounts.zoho.com	https://www.zohoapis.com	! Zoho CRM documentation — not ERP-documented
EU	https://accounts.zoho.eu	https://www.zohoapis.eu	! Zoho CRM documentation — not ERP-documented
AU	https://accounts.zoho.com.au	https://www.zohoapis.com.au	! Zoho CRM documentation — not ERP-documented
JP	https://accounts.zoho.jp	https://www.zohoapis.jp	! Zoho CRM documentation — not ERP-documented
CA	https://accounts.zohocloud.ca	https://www.zohoapis.ca	! Zoho CRM documentation — not ERP-documented

OAuth scopes — least privilege, derived from the specification

Missing scopes:ERP.fixedasset.CREATE. Re-authorise the connection and grant them.XISG-ZNT-001

Scope	Modules	Ops	Business impact if missing	Granted
ERP.accountants.CREATE	chart-of-accounts, journals	21	CWIP adjustment and capitalisation transfer cannot be posted.	✓ Granted
ERP.accountants.DELETE	chart-of-accounts, journals	9	CWIP adjustment and capitalisation transfer cannot be posted.	✓ Granted
ERP.accountants.READ	chart-of-accounts, journals	10	CWIP adjustment and capitalisation transfer cannot be posted.	✓ Granted
ERP.accountants.UPDATE	chart-of-accounts, journals	3	CWIP adjustment and capitalisation transfer cannot be posted.	✓ Granted
ERP.bills.CREATE	bills	8	Actual CWIP cannot be read. Commitment never converts to actual.	✓ Granted
ERP.bills.DELETE	bills	4	Actual CWIP cannot be read. Commitment never converts to actual.	✓ Granted
ERP.bills.READ	bills	6	Actual CWIP cannot be read. Commitment never converts to actual.	✓ Granted
ERP.bills.UPDATE	bills	4	Actual CWIP cannot be read. Commitment never converts to actual.	✓ Granted
ERP.contacts.CREATE	contacts	9	Master data for contacts cannot be synchronised.	✓ Granted
ERP.contacts.DELETE	contacts	2	Master data for contacts cannot be synchronised.	✓ Granted
ERP.contacts.READ	contacts	7	Master data for contacts cannot be synchronised.	✓ Granted
ERP.contacts.UPDATE	contacts	3	Master data for contacts cannot be synchronised.	✓ Granted
ERP.custommodules.ALL	custom-modules	7	Master data for custom-modules cannot be synchronised.	✓ Granted

- Connection profile per legal entity and environment
- Scopes derived from Zoho's specification, not from the documentation page
- Module-by-module connectivity console with endpoint, scope and result
- Secrets stored as references and never displayed after saving

869 operations, taken from Zoho's own specification

Integration scope is evidenced rather than assumed, which removes a whole class of build risk.

CAPEX & WBS Control Hub

All entities

All plants

Compact

Approvals 0

Alerts 0

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WORK

Executive Dashboard

My Approvals

Alerts & Exceptions

PROJECT CONTROL

CAPEX Projects

WBS Explorer

Budget Planning Grid

Budget Availability Check

Budget Revisions

PROCUREMENT & ACTUALS

Purchase Requests

Commitments (PO)

GRN & Receipts

Vendor Bills & CWIP

Commitment Reconciliation

CLOSURE

Capitalisation

INTEGRATION

Zoho ERP Connector

API Inventory

GOVERNANCE

Audit Trail

Home > Integration > API Inventory

Zoho ERP API Inventory

Operations 869

Modules 78

On /erp/v3 776

On /erp/v1 93

Generated mechanically from Zoho's own published OpenAPI bundle (sha256 E95A039969F5C2FC...

The ERP API is not uniformly /erp/v3.93 operations across 14 modules are served from /erp/v1, so the connector resolves the base path per module rather than per connection.0A5-81

Module	Business object	API	Operations	OAuth scopes	Needed
addons	Addons	v3	10	ERP.addons.CREATE, ERP.addons.DELETE, ERP.addons.READ, ERP.addons.UPDATE	
bank-accounts	Bank Accounts	v3	10	ERP.banking.CREATE, ERP.banking.DELETE, ERP.banking.READ, ERP.banking.UPDATE	
bank-rules	Bank Rules	v3	5	ERP.banking.CREATE, ERP.banking.DELETE, ERP.banking.READ, ERP.banking.UPDATE	
bank-transactions	Bank Transactions	v3	19	ERP.banking.CREATE, ERP.banking.DELETE, ERP.banking.READ, ERP.banking.UPDATE	
base-currency-adjustment	Base Currency Adjustment	v3	8	ERP.accountants.CREATE, ERP.accountants.DELETE, ERP.accountants.READ	
bill-of-materials	Bill of Materials	v3	8	ERP.manufacturing.CREATE, ERP.manufacturing.DELETE, ERP.manufacturing.READ, ERP.manufacturing.UPDATE	
bills	Bills	v3	22	ERP.bills.CREATE, ERP.bills.DELETE, ERP.bills.READ, ERP.bills.UPDATE	✓ Yes
cards	Cards	v3	3	ERP.customers.DELETE, ERP.customers.READ	
chart-of-accounts	Chart Of Accounts	v3	12	ERP.accountants.CREATE, ERP.accountants.DELETE, ERP.accountants.READ, ERP.accountants.UPDATE	✓ Yes
compositeitems	Composite Items	v3	11	ERP.compositeitems.CREATE, ERP.compositeitems.DELETE, ERP.compositeitems.READ, ERP.compositeitems.UPDATE	
contact-persons	Contact Persons	v3	6	ERP.contacts.CREATE, ERP.contacts.DELETE, ERP.contacts.READ, ERP.contacts.UPDATE	
contacts	Contacts	v3	21	ERP.contacts.CREATE, ERP.contacts.DELETE, ERP.contacts.READ, ERP.contacts.UPDATE	✓ Yes
cost-templates	Cost Templates	v3	5	ERP.manufacturing.CREATE, ERP.manufacturing.DELETE, ERP.manufacturing.READ, ERP.manufacturing.UPDATE	
coupons	Coupons	v3	11	ERP.coupons.CREATE, ERP.coupons.DELETE, ERP.coupons.READ, ERP.coupons.UPDATE	
credit-notes	Credit Notes	v3	30	ERP.creditnotes.CREATE, ERP.creditnotes.DELETE, ERP.creditnotes.READ, ERP.creditnotes.UPDATE, ERP.invoices.UPDATE	
currency	Currency	v3	10	ERP.settings.CREATE, ERP.settings.DELETE, ERP.settings.READ, ERP.settings.UPDATE	
custom-modules	Custom Modules	v3	12	ERP.custommodules.ALL, ERP.settings.CREATE, ERP.settings.DELETE, ERP.settings.READ, ERP.settings.UPDATE	✓ Yes
customer-debit-notes	Customer Debit Notes	v3	5	ERP.invoices.CREATE, ERP.invoices.DELETE, ERP.invoices.READ, ERP.invoices.UPDATE	
customer-payments	Customer Payments	v3	13	ERP.customerpayments.CREATE, ERP.customerpayments.DELETE, ERP.customerpayments.READ, ERP.customerpayments.UPDATE	
delivery-challans	Delivery Challans	v3	17	ERP.invoices.CREATE, ERP.invoices.DELETE, ERP.invoices.READ, ERP.invoices.UPDATE	
departments	Departments	v1	5	ERP.department.CREATE, ERP.department.DELETE, ERP.department.READ, ERP.department.UPDATE	✓ Yes
designations	Designations	v1	5	ERP.designation.CREATE, ERP.designation.DELETE, ERP.designation.READ, ERP.designation.UPDATE	
employee-salary	Employee Salary	v1	9	ERP.salary.CREATE, ERP.salary.DELETE, ERP.salary.READ, ERP.salary.UPDATE	
employee-statutory	Employee Statutory	v1	10	ERP.employee.CREATE, ERP.employee.READ, ERP.employee.UPDATE	
employees	Employees	v1	11	ERP.employee.CREATE, ERP.employee.DELETE, ERP.employee.READ, ERP.employee.UPDATE	
estimates	Estimates	v3	25	ERP.estimates.CREATE, ERP.estimates.DELETE, ERP.estimates.READ, ERP.estimates.UPDATE	

- 78 modules and 163 OAuth scopes, machine-extracted, not hand-written
- The connector cannot reference an endpoint that does not exist
- 14 modules are served from a different API version, and are flagged
- Each row carries its verification date

Who did what, when, and why

Maker-checker evidence is a by-product of using the system, not a separate exercise.

CAPEX & WBS Control Hub

All entities

All plants

Compact

Approvals 0

Alerts 0

NR

WORK

Executive Dashboard

My Approvals

Alerts & Exceptions

PROJECT CONTROL

CAPEX Projects

WBS Explorer

Budget Planning Grid

Budget Availability Check

Budget Revisions

PROCUREMENT & ACTUALS

Home > Audit Trail

Audit Trail

Entries 4

Complete maker-checker audit trail

At	Actor	Action	Object	Reference	Detail
2026-08-06T04:30:40	U-ADM	OAuth_Authorised	ZohoConnection	CONN-01	44 of 45 required scopes granted. Secrets stored as references only.
2026-08-06T03:20:55	U-ADM	OAuth_Authorised	ZohoConnection	CONN-01	43 of 45 required scopes granted. Secrets stored as references only.
2026-08-06T03:20:38	U-REQ	PR_CREATED	PurchaseRequest	PR-021	PR-2026-0021 for CAPEX-2026-001.03.01 value ₹25,00,000. Budget check: EXCEEDS_BUDGET.
2026-08-06T09:00:00	U-ADM	SEED	System	-	POC dataset loaded: 2 entities, 2 plants, 3 projects, 3-level WBS, 10 budget heads, 20 PRs, 15 POs, partial GRNs, partial bills, PO amendment, PO cancellation, PO closure with residual, budget supplement, budget transfer, one overrun, one capitalisation case, one abandoned WBS.

- Every approval, amendment, cancellation and capitalisation recorded

● Exception reasons captured against the approver's name

● Budget revisions retain reason, approver, date and approval reference

● Drill-down from a summary figure to the underlying document

A purchase order line contributes to commitment only for its unbilled balance.

Billing moves value from commitment into actual cost. It is never added to both, so exposure does not move as billing progresses.

Transaction position	PO value	Actual bill	Open commitment	Budget exposure
Purchase order approved, not billed	₹20L	₹0	₹20L	₹20L
Partially billed	₹20L	₹12L	₹8L	₹20L
Fully billed	₹20L	₹20L	₹0	₹20L
Cancelled before billing	₹20L	₹0	₹0	₹0

Nothing falls between the two

Goods can be received well before the vendor bills. That value is tracked as a separate received-but-unbilled bucket and reported as exposure, so it is never lost between commitment and actual cost.

Budget is tested where it is approved

Spending posts on a work package while budget sits above it. The check resolves to the nearest element that carries budget and evaluates its whole subtree, so a parent's budget cannot look untouched while its children consume it.

VERIFIED IN THE RUNNING APPLICATION

PO-2026-0003 carries ₹17,40,000 billed plus ₹17,00,000 open commitment against ₹34,40,000 ordered. Its exposure is unchanged as billing progresses. A second control tracks received-but-unbilled value separately, so nothing can fall between commitment and actual cost.

We checked rather than assumed. Some of what we found changes the plan.

Works today

- Purchase order lines carry project, tags and custom fields, so CAPEX and WBS coding can ride on the line
- Bill lines link back to the purchase order line, so commitment converts to actual without double counting
- Journals can be coded at line level for CWIP transfer and capitalisation
- Vendors, items, locations, taxes, currencies and chart of accounts all synchronise

Needs a decision or a workaround

- Zoho ERP has no Purchase Request API. The request stays in the Control Hub
- Goods receipt lines carry no link to the purchase order line, and cannot be listed. Unbilled receipt value is reconstructed here
- Custom fields cannot be created through the API. An administrator must create them in Zoho first
- Fixed assets cannot be synchronised incrementally, so asset refresh is a full reload

None of this is guesswork. Every statement above traces to Zoho's own published specification.

869 operations across 78 modules were extracted mechanically and checked, which is why the limitations surfaced now rather than during the build.

ACTION REQUIRED BEFORE A LIVE CONNECTION

A Zoho administrator must create the CAPEX code, WBS code and budget head custom fields in the Zoho interface. There is no API that can do this. Every subsequent synchronisation depends on it, so it is worth starting now.

Demonstrated and working

- The complete control loop, from project creation to asset capitalisation
- Three-level work breakdown structure with figures that roll up correctly
- Budget availability enforced before commitment, at the right level of the hierarchy
- Commitment converted to actual cost without double counting, proven line by line
- Budget revision with the original budget preserved intact
- Exception approval that cannot proceed without a recorded reason
- Capitalisation gated on a complete allocation of the CWIP balance
- Zoho connector grounded in the vendor's own API specification
- Complete audit trail across every action

Not yet built, and deliberately so

- User sign-on and enforcement of the role model
- A live Zoho connection. The connector runs against the specification in simulation
- Production database with locking, so two approvers cannot consume the same remaining budget
- Migration of existing CWIP balances and open commitments
- Automated regression testing
- The approval matrix itself: amounts, roles and escalation are placeholders until Atha Group confirms them

1

Walk the application together

See the full loop with your project control and finance teams, on your own terminology and figures.

2

Confirm the control decisions

Should an approved purchase request reserve budget before a PO exists? What are the approval thresholds by amount and role?

3

Open the Zoho prerequisites

Administrator access to create the CAPEX, WBS and budget head fields, and confirmation of the CWIP ledger structure.

4

Agree the pilot scope

One plant and one live project is enough to prove the control end to end before wider rollout.

Commercials and timelines are not included in this pack and will be quoted separately once scope is agreed.

Control the commitment, not just the invoice.

The proof of concept is running and can be demonstrated end to end today.
Everything shown in this pack is a real screen from the working application.

